Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… ☐ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. ☐ Read or plot how pH links to rate. Say only what the data supports. ☐ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Word page

Enzyme and substrate

active site — The part of an enzyme where the substrate binds. The shapes fit each other.

denaturation — A change in enzyme shape. This stops the active site binding as it should.

diffusion — Net movement of particles. They move from higher to lower concentration.

digestion — A chemical change. It breaks large molecules down into smaller molecules.

substrate — The substance that an enzyme acts on.

enzyme — A biological catalyst. It speeds up a reaction. It is not used up.

specificity — An enzyme acts on particular substrates. They fit its active site.

Measure and compare

relative rate — One divided by the endpoint time. Match the starting amounts. Use the same endpoint.

reciprocal — One divided by a number.

pH — A measure of how acid or alkaline a substance is.

endpoint — The agreed point we can see and record. We record the time when it is reached.

control variable — A factor we keep the same. This makes the comparison fair.

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Word page · continued

Diagram

Dark dots show the first-series rates. The graph uses the supplied data; it is not a record of your own practical.

Relative rate = 1/time. For 120 s: 1/120 = 0.00833 s^{-1}. For 90 s: 1/90 = 0.0111 s^{-1}. A shorter endpoint time gives a larger reciprocal, for matched amounts and the same stopping rule.

